



PAN3029 470-510MHz

TEST REPORT

V1.0 Dec. 2023

Shanghai Panchip Microelectronics Co., Ltd.

PAN3029/PAN 3060 Series

Table of contents

Table of contents	2
1 Overview	4
2 Test Summary	5
2.1 Test Content	5
2.2 Test Object	5
2.3 Test Equipment	6
3 Test Content	7
3.1 Receiving Performance Test	7
3.1.1 Receiving Sensitivity Test	7
3.1.1.1 Test Environment	7
3.1.1.2 Test Steps	7
3.1.1.3 Test Results	7
Sensitivity of Different Modes at Different Packet Lengths	7
Sensitivity of Different Frequency Points	8
3.1.2 Receiving Current Test	9
3.1.2.1 Test Environment	9
3.1.2.2 Test Steps	9
3.1.2.3 Test Results	10
3.2 Transmitting Performance Test	11
3.2.1 Transmitting Power and Current Test	11
3.2.1.1 Test Environment	11
3.2.1.2 Test Steps	11
3.2.1.3 Test Results	11
Transmitting Power and Current at Each Level	12
Transmitting Power and Current at Different Frequency Points	12
3.2.2 Transmitting Harmonic Test	13
3.2.2.1 Test Environment	13
3.2.2.2 Test steps	13
3.2.2.3 Test results	14
3.2.3 Transmit bandwidth test	15
3.2.3.1 Test environment	15
3.2.3.2 Test steps	15
3.2.3.3 Test results	15
3.3 Low power performance test	18
3.3.1 Deepsleep&Sleep current test	18
3.3.1.1 Test environment	18
3.3.1.2 Test steps	18
3.3.1.3 Test results	18
3.4 Distance test in various environments	19
3.4.1 Ground communication in Suzhou city	19
3.4.1.1 Test environment	19
3.4.1.2 Test steps	22
3.4.1.3 Test results	22
3.4.2 Communication between rooftop and ground in Suzhou city	24
3.4.2.1 Test environment	24
3.4.2.2 Test steps	28

3.4.2.3 Test results	28
3.4.3 Communication by Taihu Lake in Suzhou suburbs.....	29
3.4.3.1 Test environment.....	29
3.4.3.2 Test steps	39
3.4.3.3 Test results.....	39
3.4.4 Suzhou Zhongxin Building wall penetration test.....	41
3.4.4.1 Test environment.....	41
3.4.4.2 Test steps	42
3.4.4.3 Test results	42
Revision history	44
Contact information	45

1 Overview

This report is based on the test data of the PAN3029 evaluation board PAN3029EVB, and is intended to help customers understand the index parameters of the PAN3029 evaluation board. Since the switch combiner is added to the evaluation board, there will be a loss of about 0.3dB in power and sensitivity. At the same time, the test methods of some parameters of the evaluation board are different from the product manual, so the test results of some parameters will be slightly different from the index parameters in the product manual.

References and links:

- "PAN3029 Product Manual"
- "PAN3029 Evaluation Board User Manual"
- <https://bbs.panchip.com/forum.php?mod=viewthread&tid=8256&extra=page%3D1>

2 Test Summary

2.1 Test Content

The test contents mainly include the following aspects:

- Receiving performance test
- Transmitting performance test
- Power consumption test
- Distance test

2.2 Test Object

The entire test is based on the PAN3029EVB evaluation board, mainly testing 3 chips, and the number of tests for different projects will vary.

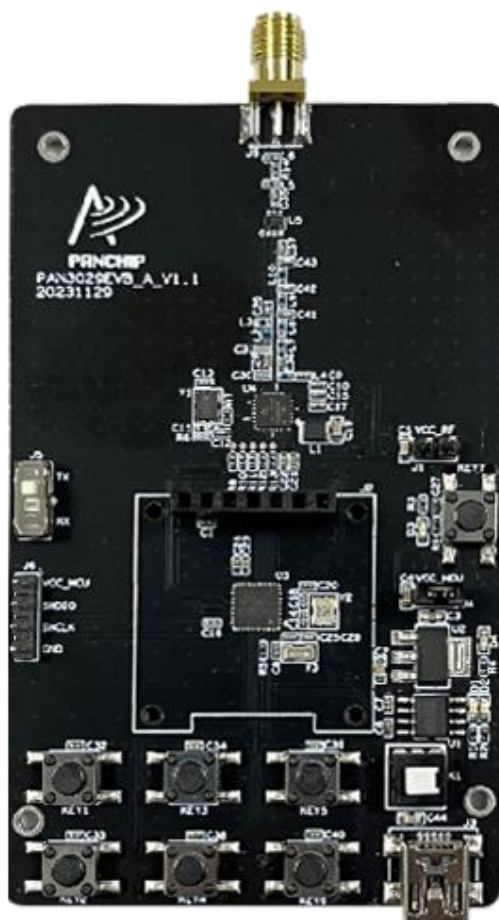


Figure 2-1 PAN3029EVB evaluation board

2.3 Test Equipment

Table 2-1 Test equipment

No.	Device Name	Device Model	Quantity
1	Spectrum analyzer	N9020B	1
2	Signal generator	N5182B/E4438C	2
3	Digital multimeter	34465A	1
4	Network analyzer	ZVB8	1
5	Shielded box	GX-5935A	1
6	USB to serial cable	/	1
7	RF coaxial cable and connector	/	1
8	Antenna	470-510MHz	2

3 Test Content

3.1 Receiving Performance Test

3.1.1 Receiving Sensitivity Test

3.1.1.1 Test Environment



Figure 3-1 Sensitivity test environment

3.1.1.2 Test Steps

1. Set up the environment according to Figure 3-1;
2. Use the button to make PAN3029 enter the LDO mode receiving state;
3. Use the button to set the parameters BW/SF/frequency/Payload Length;
4. Adjust the output power of the signal source and record the power that meets the PER requirement.

3.1.1.3 Test Results

Sensitivity of Different Modes at Different Packet Lengths

The test results are shown in the following tables:

Table 3-1 Receiver sensitivity data (PER=1%) with switch insertion loss compensated

CodingRate	Payload Length(Bytes)	BW(KHz)	SF	Sensitivity(dBm)		
				Chip #1	Chip #2	Chip #3
CR1	64	125	5	-118	-117	-118
			6	-120	-120	-120
			7	-124	-123	-124
			8	-126	-126	-127
			9	-129	-129	-129
			10	-132	-132	-132
			11	-135	-134	-135
			12	-137	-137	-137
		62.5	12	-141	-140	-141

Table 3-2 Receiver sensitivity data (PER=5%) with switch insertion loss compensated

CodingRate	Payload Length(Bytes)	BW(KHz)	SF	Sensitivity(dBm)		
				Chip #1	Chip #2	Chip #3
CR1	64	125	5	-118	-118	-118
			6	-120	-121	-121
			7	-124	-124	-125
			8	-127	-127	-128
			9	-130	-130	-130
			10	-133	-132	-133
			11	-135	-135	-136
			12	-137	-138	-138
		62.5	12	-141	-141	-141

Sensitivity of Different Frequency Points

The test results are shown in the following table:

Table 3-3 Receiving sensitivity data at different frequencies (PER=5%) with switch insertion loss compensated

Frequency (MHz)	CodingRate	Payload Length(Bytes)	BW(KHz)	SF	Sensitivity(dBm)		
					Chip #1	Chip #2	Chip #3
470	CR1	64	125	9	-129	-129	-130
478					-130	-130	-130
486					-129	-130	-129
494					-129	-129	-130
502					-129	-129	-130
510					-129	-129	-129

3.1.2 Receiving Current Test

3.1.2.1 Test Environment



Figure 3-2 Receive current test environment

3.1.2.2 Test Steps

1. Set up the environment according to Figure 3-2;
2. Use the buttons to make PAN3029 enter LDO mode & DCDC mode;
3. Use the buttons to set the parameters BW/SF/frequency point;
4. Record the RX current of different frequencies and different modes with the digital multimeter.

3.1.2.3 Test Results

The test results are shown in the following table:

Table 3-4 Receiving current at different frequencies (LDO mode & DCDC mode)

Frequency (MHz)	BW(KHz)	SF	LDO mode RX current (mA)			DCDC mode RX current (mA)		
			Chip #1	Chip #2	Chip #3	Chip #1	Chip #2	Chip #3
470	125	9	7.98	7.82	7.86	4.31	4.4	4.34
478			8	7.84	7.88	4.32	4.42	4.35
486			8.02	7.86	7.89	4.33	4.43	4.36
494			8.04	7.87	7.91	4.34	4.44	4.37
502			8.06	7.91	7.94	4.35	4.45	4.38
510			8.09	7.93	7.97	4.36	4.46	4.39

3.2 Transmitting Performance Test

3.2.1 Transmitting Power and Current Test

3.2.1.1 Test Environment

As shown in Figure 3-3.

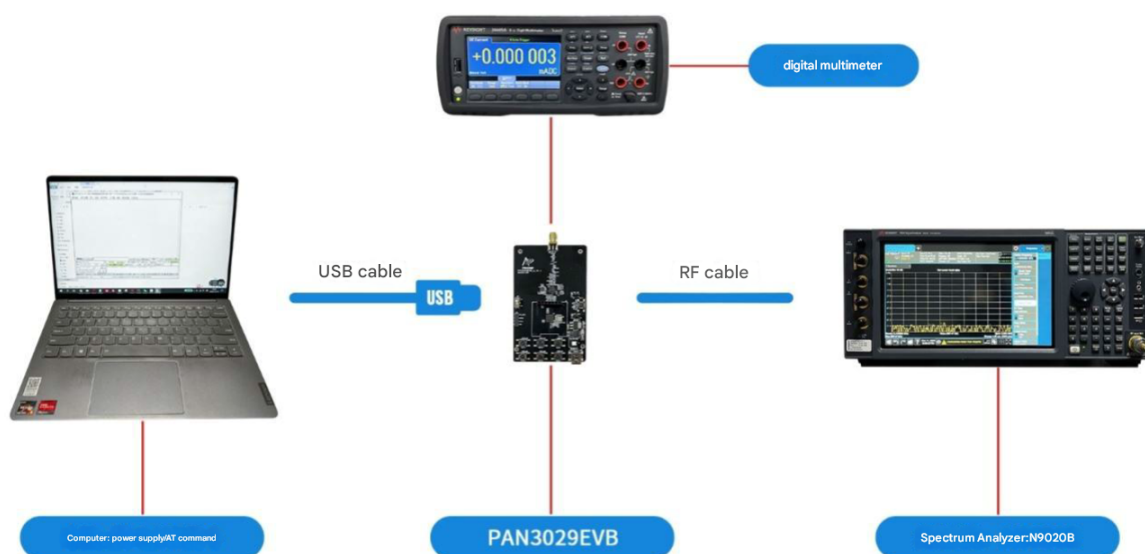


Figure 3-3 Transmit power and current test environment

3.2.1.2 Test Steps

1. Set up the environment according to Figure 3-3;
2. Use the button to make PAN3029 enter the carrier transmission mode;
3. Use the button to set the power of PAN3029;
4. The spectrum analyzer records the transmission power, and the digital multimeter records the transmission current.

3.2.1.3 Test Results

The test results are shown in the following table:

Transmitting Power and Current at Each Level

Table 3-5 Transmitting power and current at each level

Frequency (MHz)	Level	Chip #3	
		Power (dBm)	Current (mA)
490	23	19.90	97.55
	22	19.54	94.24
	21	18.34	80.84
	20	17.47	67.83
	19	16.33	60.04
	18	15.34	53.77
	17	13.71	45.52
	16	13.52	44.63
	15	13.01	42.77
	14	12.31	40.28
	13	11.20	36.76
	12	10.48	34.70
	11	7.92	26.82
	10	7.38	25.76
	9	4.51	20.50
	8	4.10	19.89
	7	2.40	17.73
	6	-1.08	14.65
	5	-1.51	14.05
	4	-2.33	13.45
	3	-5.80	12.14
	2	-17.65	8.02
	1	-26.67	7.25

Transmitting Power and Current at Different Frequency Points

Table 3-6 Transmitting power and current at different frequencies

Frequency (MHz)	Level	Chip #1		Chip #2		Chip #3	
		Power (dBm)	Current (mA)	Power (dBm)	Current (mA)	Power (dBm)	Current (mA)
470	23	20.8	117	19.7	97	20.25	108
	20	18.6	83.1	16.58	64	17.8	73
	15	14.87	55.8	12	39.3	14	49.7
	12	11.81	42.9	9.75	33.5	11.05	39.5
478	23	20.7	109.8	19.83	96.9	20.3	104.7
	20	18.9	82.85	17	63.7	18.17	74.4
	15	15.1	55.3	12.4	40	14.3	50.1
	12	11.8	41.1	10.12	33.6	11.3	38.9
486	23	20.2	99.4	19.7	92	20.04	97.3
	20	18.7	79	17.2	63.1	18.2	72.4
	15	15	52.5	12.7	39.7	14.4	48.7
	12	11.42	38.1	10.15	32.7	11.1	36.9

494	23	19.8	92.8	19.43	87.5	19.7	91.2
	20	18.57	75.2	17.2	61.7	18.05	69.3
	15	14.75	50	12.7	38.9	14.26	46.6
	12	11.1	36	10	31.6	11.27	38.9
502	23	19.56	89.9	19.34	86.8	19.57	89.8
	20	18.43	73.6	17.17	61.6	18	68.7
	15	14.6	48.9	12.7	39	14.16	46.1
	12	10.9	35.2	9.96	31.6	10.71	34.7
510	23	19.33	87.1	19.36	87.7	19.5	90
	20	17.34	62.4	17.22	63	18	69.8
	15	14.5	48.3	12.65	39.5	14.2	46.8
	12	10.67	34.5	10	31.9	10.7	35

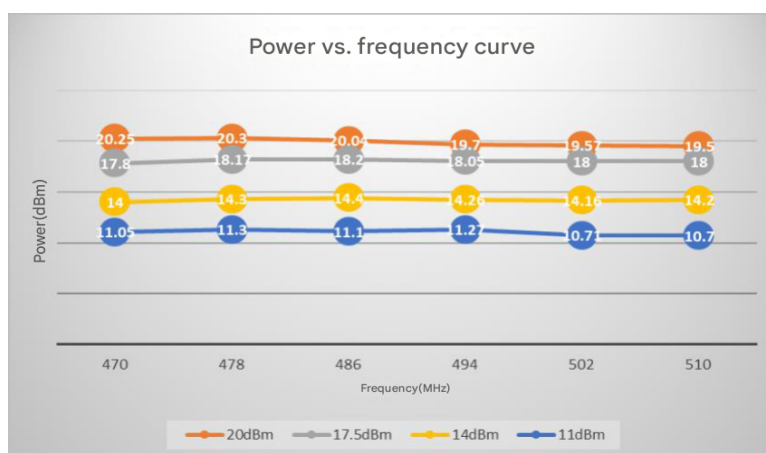


Figure 3-4 Power changes with frequency at several power levels

3.2.2 Transmitting Harmonic Test

3.2.2.1 Test Environment

As shown in Figure 3-3.

3.2.2.2 Test steps

1. Set up the environment according to Figure 3-3;
2. Use the button to make PAN3029 enter the carrier transmission mode;
3. Use the button to set the power and frequency of PAN3029;
4. Record the power and harmonics on the spectrum analyzer.

3.2.2.3 Test results

The test results are shown in the following table:

Table 3-7 Power of second harmonic and third harmonic at maximum transmission power at different frequencies

Frequency (MHz)	Level	Chip #1			Chip #2			Chip #3		
		Power (dBm)	Second Harmonic (dBm)	Third harmonic (dBm)	Power (dBm)	Second Harmonic (dBm)	Third harmonic (dBm)	Power (dBm)	Second Harmonic (dBm)	Third harmonic (dBm)
470	23	20.8	-41	-41	19.7	-43	-44.2	20.25	-41.5	-48
478	23	20.7	-42.7	-43.2	19.83	-43	-46.2	20.3	-42.8	-48.3
486	23	20.2	-45	-45.6	19.7	-44.5	-47.5	20.04	-44	-48.4
494	23	19.8	-44	-47.6	19.43	-44	-49	19.7	-43.7	-48.5
502	23	19.56	-43	-48.6	19.34	-43.7	-50	19.57	-43	-48.5
510	23	19.33	-43	-48.9	19.36	-43	-49	19.5	-42.5	-48.6

The test screenshot is shown in Figure 3-5.

Mark1: Carrier frequency signal

Mark2: Second harmonic

Mark3: Third harmonic

Mark4: Fourth harmonic

Mark5: Fifth harmonic

Mark6: Sixth harmonic

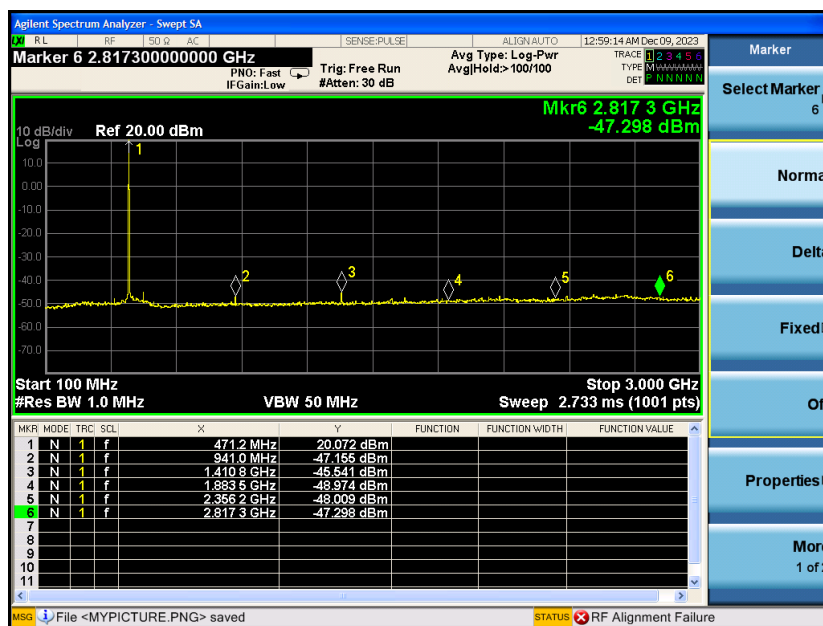


Figure 3-5 Emitted harmonics

3.2.3 Transmit bandwidth test

3.2.3.1 Test environment

As shown in Figure 3-3.

3.2.3.2 Test steps

1. Set up the environment according to Figure 3-3;
2. Use the button to make PAN3029 enter the normal transmission mode;
3. Use the button to set the BW/SF/frequency of PAN3029;
4. Record the broadband on the spectrum analyzer.

3.2.3.3 Test results

The test results are as follows:

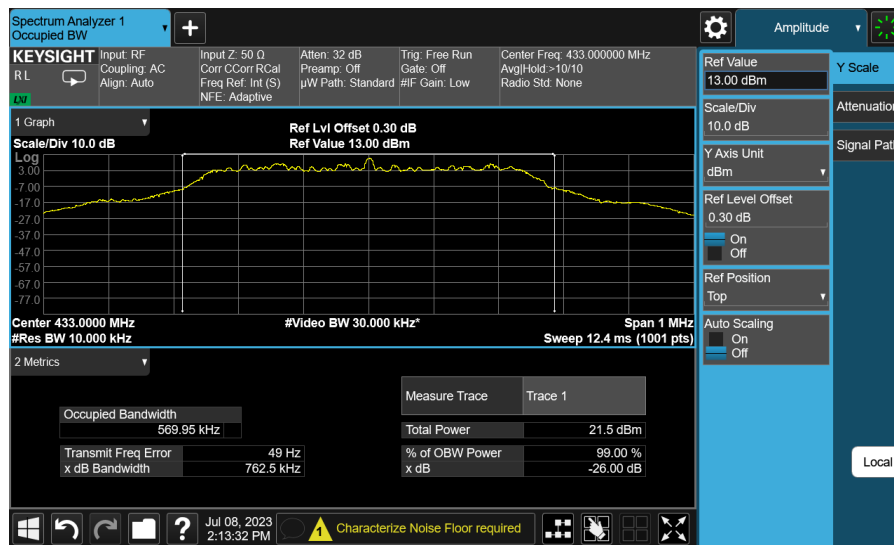


Figure 3-6 BW= 500K bandwidth

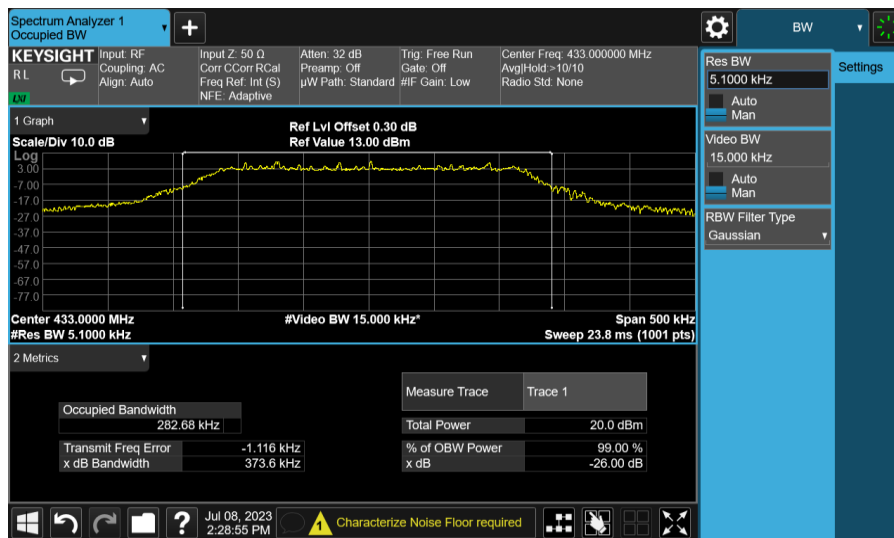


Figure 3-7 BW= 250K bandwidth

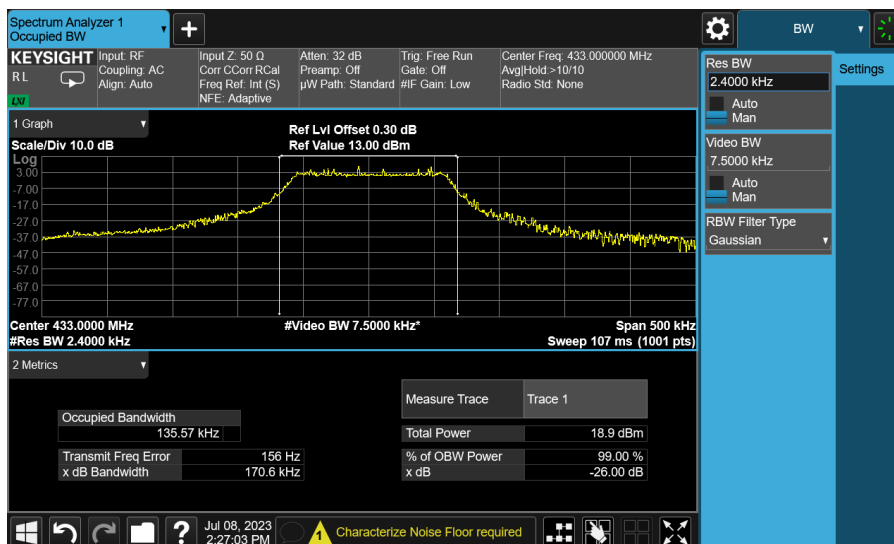


Figure 3-8 BW= 125K bandwidth

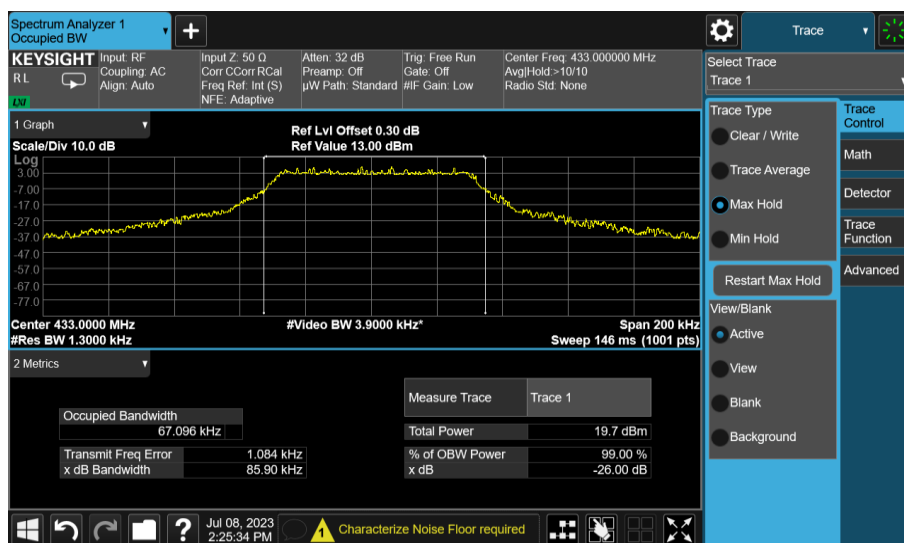


Figure 3-9 BW= 62.5K bandwidth

3.3 Low power performance test

3.3.1 DeepSleep&Sleep current test

3.3.1.1 Test environment

As shown in Figure 3-10.

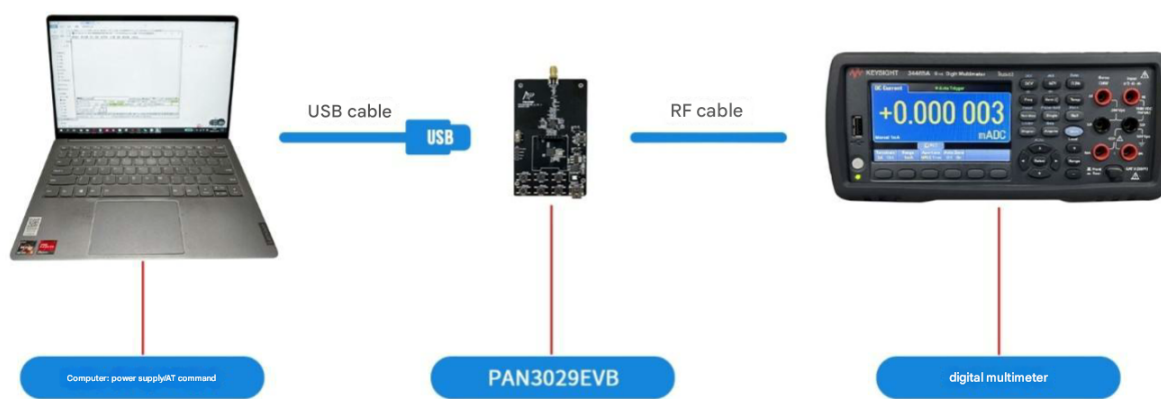


Figure 3-10 Low power consumption current test environment

3.3.1.2 Test steps

1. Set up the environment according to Figure 3-10;
2. Use the button to make PAN3029 enter the low power performance mode (DeepSleep&Sleep);
3. Record the current of the digital multimeter.

3.3.1.3 Test results

The test results are shown in the following table:

Table 3-8 DeepSleep and Sleep current

Chip	DeepSleep current (nA)	Sleep current (nA)
#1	260	621
#2	280	643
#3	245	619

3.4 Distance test in various environments

A total of four scenarios were tested, including ground-to-ground test in the urban area, rooftop-to-ground test in the urban area, lakeside-to-lake test in the suburbs, and indoor wall-to-wall test.

3.4.1 Ground communication in Suzhou city

3.4.1.1 Test environment

On Chuangyuan Road in Suzhou City, there are fewer obstacles in the urban area, more traffic lights, and more motor vehicles on the road. Map screenshots of different distances will be provided later.



Figure 3-11 Suzhou ground pull distance test environment a

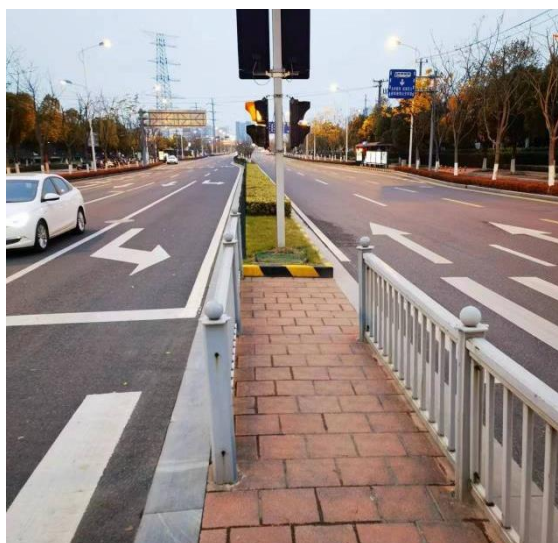


Figure 3-12 Suzhou ground pull distance test environment b

1. Module status:

- 1) Transmitter: placed in hand, 1.3m above the ground

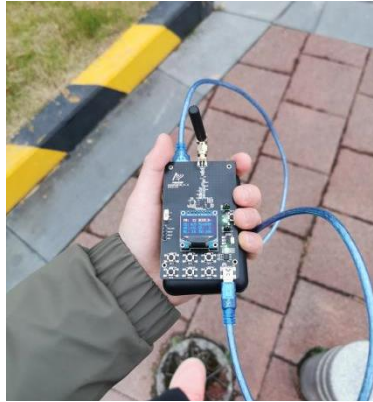


Figure 3-13 Suzhou ground distance TX end

- 2) Receiver: placed in hand, 1.3m above the ground



Figure 3-14 Suzhou ground distance RX end

2. Parameter configuration:

- 1) Frequency: 499MHz
- 2) Payload length: 10Bytes
- 3) SF and BW: There will be some changes at different distances, see test data for details
- 4) CR: 1
- 5) Output power: 20dBm

3. Screenshots of maps at different distances:

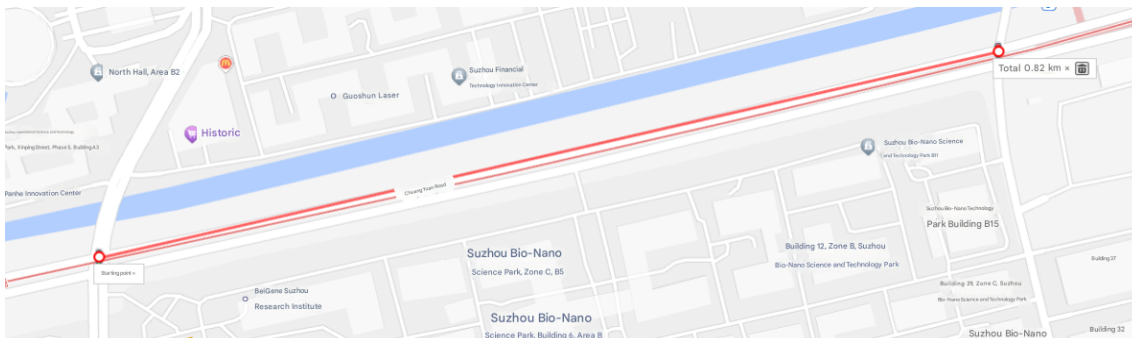


Figure 3-15 Screenshot at 820 meters

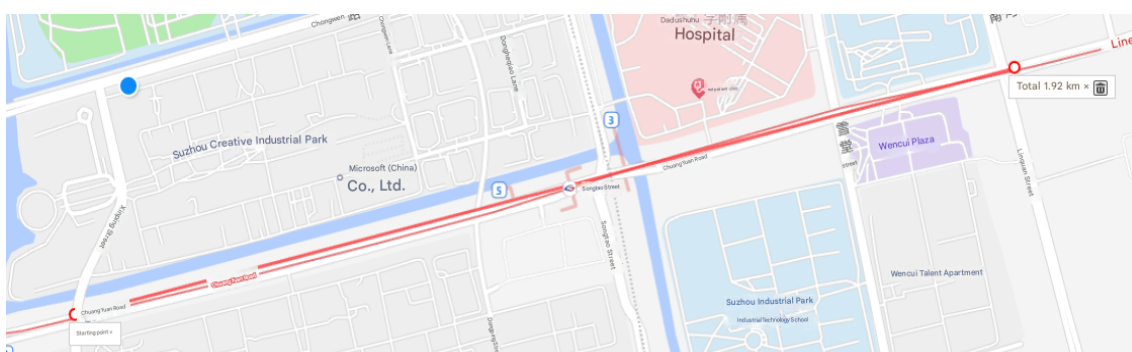


Figure 3-16 Screenshot at 1920 meters

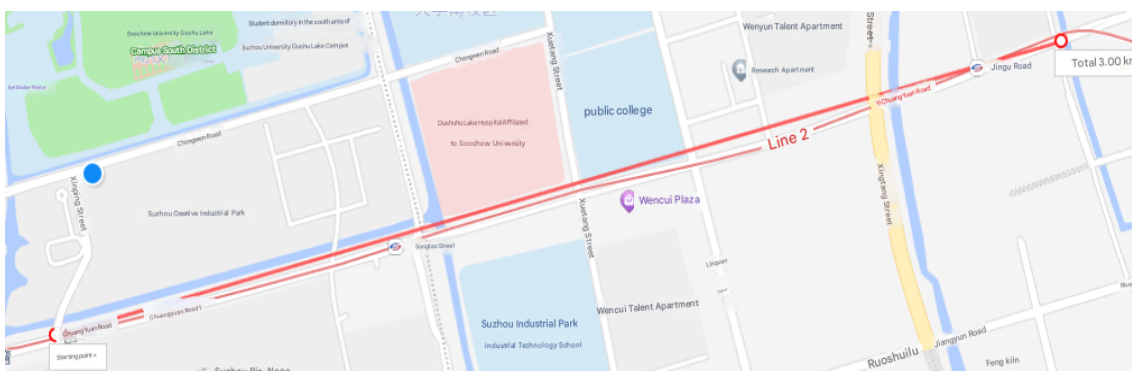


Figure 3-17 Screenshot at 3000 meters

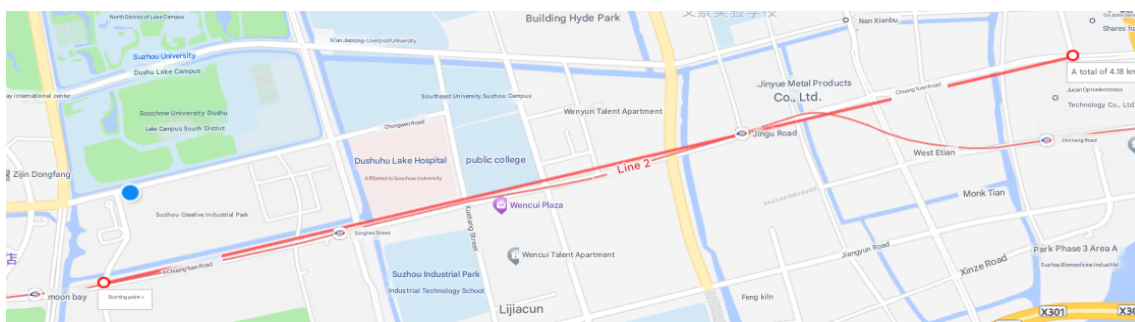


Figure 3-18 Screenshot at 4180 meters



Figure 3-19 Screenshot at 5130 meters

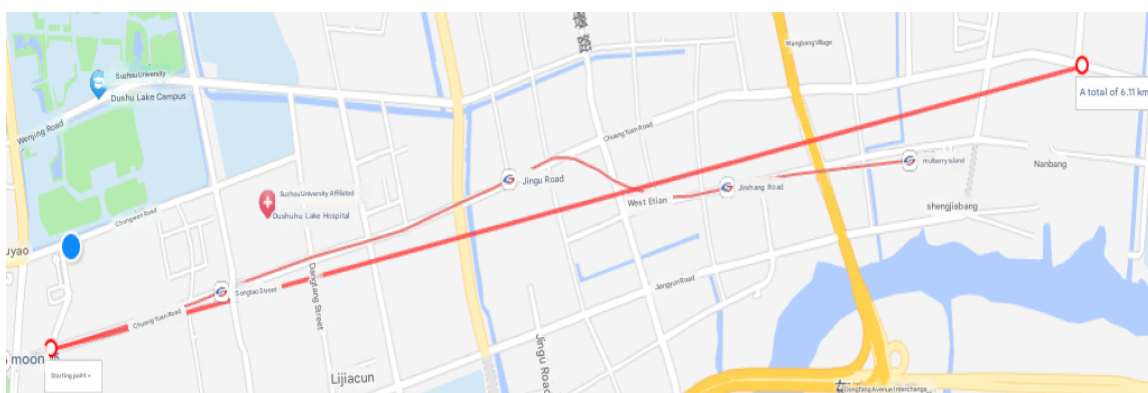


Figure 3-20 Screenshot at 6110 meters

3.4.1.2 Test steps

1. Set the TX mode and RX mode respectively through the buttons, and configure the corresponding CR/BW/SF parameters;
2. The TX end sends 100 packets of data each time through the B mode, and the RX end counts the number of packets received at different distances and different parameters.

3.4.1.3 Test results

The test results are shown in the table below. The road limit distance is 6110m, SF 12 can receive packets; 125K SF 9, 5130m can receive packets. The distance is above 4180, and the directionality is very sensitive.

The data in the table are the data with better packet reception.

Table 3-9 Ground pull distance results of Suzhou Chongwen Road

BW (kHz)	SF	Tx end position	Rx end position	Distance (m)	Packet receiving statistics	
					PAN3029	
125	5	Intersection of Chuangyuan Road and Xiping Street	Intersection of Chuangyuan Road and Dongping Street	820	94	100
	6				99	99
	7				98	100
125	5		Intersection of Chuangyuan Road and Linqun Street	1920	17	43
	6				73	96
	7				91	83
125	7		Intersection of Chuangyuan Road and Jinhai Road	3000	0	/
	8				23	55
	9				85	62
	10				90	95
125	9		Intersection of Chuangyuan Road and Jinshang Road	4180	67	88
	10				91	70
	11				80	85
125	9		Intersection of Chuangyuan Road and Guangxian Street	5130	47	20
	10				59	12
	11				74	37
	12				88	64
125	12		Intersection of Chuangyuan Road and Changyang South Street (1.3m above ground level)	6110	0	0
125	12		Intersection of Chuangyuan Road and Changyang South Street (2.0m above ground level)	6110	40	18
62.5	12		Intersection of Chuangyuan Road and Changyang South Street	6110	57	20

3.4.2 Communication between rooftop and ground in Suzhou city

3.4.2.1 Test environment

The transmitter is located on the rooftop of the 6th floor of Fuhua Technology Building, No. 199 Chongwen Road, Suzhou City. The receiver is located far away along Chuangyuan Road. The environment is relatively complex and there are high-rise buildings blocking the way.



Figure 3-21 Suzhou Fuhua Technology Building rooftop environment



Figure 3-22 Suzhou road test environment

1. Module status:

- 1) Transmitter: placed on the rooftop wall, 20m above the ground

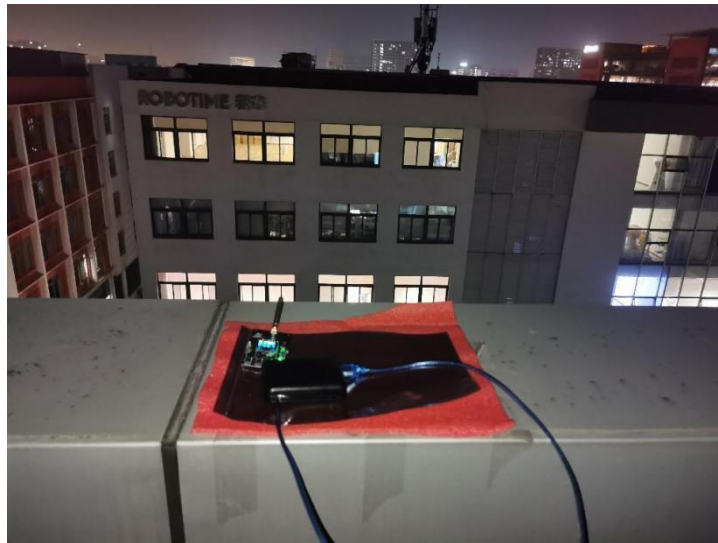


Figure 3-23 Suzhou Fuhua Technology Building rooftop TX end

2) Receiver: held in hand, 1.3m above the ground

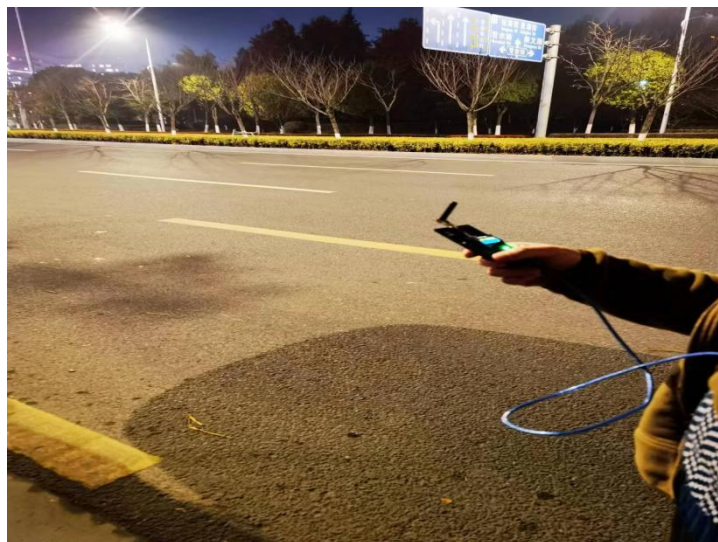
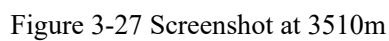
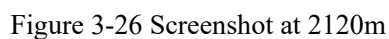
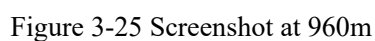


Figure 3-24 Road RX end

2. Parameter configuration:

- 1) Frequency: 499MHz
- 2) Payload length: 10Bytes
- 3) SF and BW: There will be some changes at different distances, see the test data for details
- 4) CR: 1
- 5) Output power: 20dBm

■ ■ ■



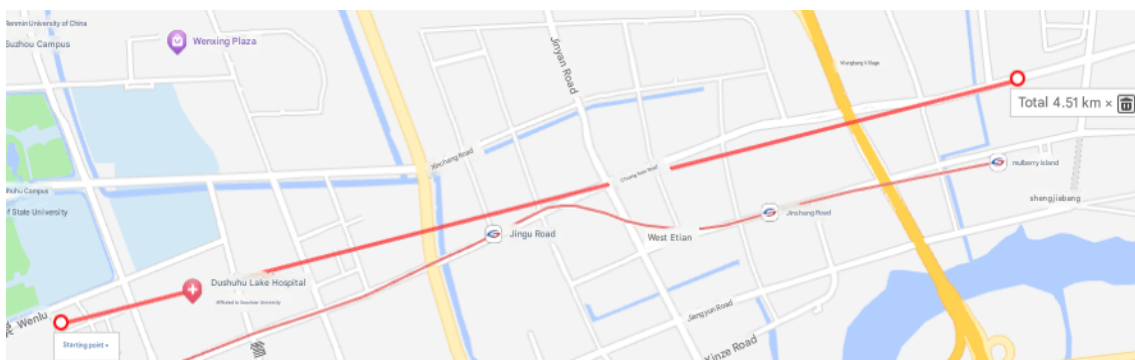


Figure 3-28 Screenshot at 4510m

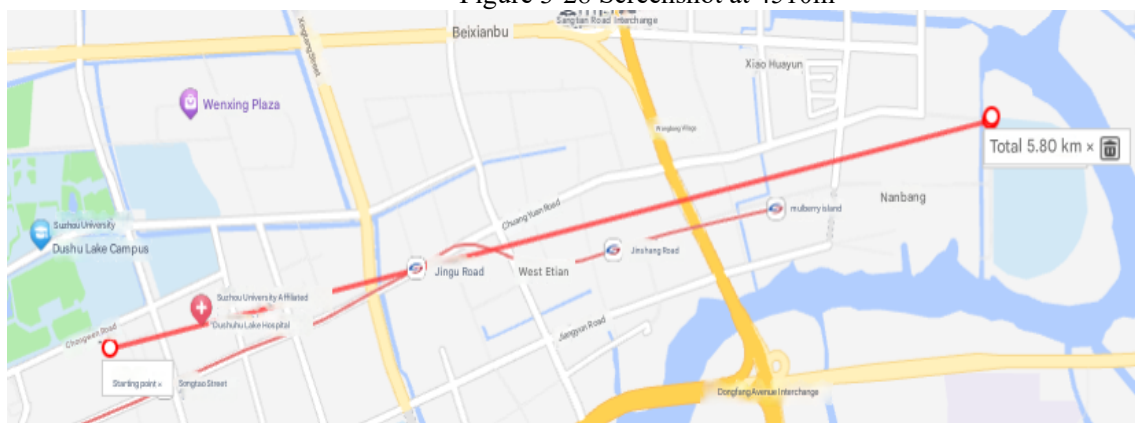


Figure 3-29 Screenshot at 5800 meters



Figure 3-30 Screenshot at 7820 meters



Figure 3-31 Screenshot at 9020 meters



Figure 3-32 Screenshot at 10000 meters

3.4.2.2 Test steps

1. Set the TX mode and RX mode respectively through the buttons, and configure the corresponding CR/BW/SF parameters;
2. The TX end sends 100 packets of data each time through B mode, and the RX end counts the number of received packets at different distances and different parameters.

3.4.2.3 Test results

The test results are shown in the table below. The extreme distance is 10km and the data packet can be received occasionally. Within 5.8km, the 250K SF12 packet reception rate reaches more than 70%. Above 5.8km, the directionality is more sensitive.

Table 3-10 Suzhou air-to-ground distance pull results

BW (kHz)	SF	Tx end position	Rx end position	Distance (m)	Packet receiving statistics	
					PAN3029	
500	5	Rooftop of Fuhua Technology Building	100m east of the intersection of Chuangyuan Road and Xuetang Street	960	39	30
250	5				99	99
125	5				100	99
	7				100	100
125	5		100m east of the intersection of Chuangyuan Road and Jingu Road	2120	4	2
	6				100	99
	7				100	99
125	7		100m east of the intersection of Chuangyuan Road and Jinshang Road	3510	25	8
	8				98	81
	9				93	98
125	8		100m east of the intersection of Chuangyuan Road and Guangxian Street	4510	0	0
	9				76	82
	10				100	100
125	9		Intersection of Chuangyuan Road	5800	23	0

	10		and Jiangyun Road		35	1
	11				16	/
	12				71	89
125	12		Intersection of Dongfang Avenue and Gangling Road (2.0m above ground level)	7820	27	42
62.5	12				21	30
125	12		Intersection of Dongfang Avenue and Xiwugu Road (2.0m above ground level)	9020	8	/
62.5	12				2	/
62.5	12		Intersection of Qingsha Road and Mingshi Road (2.0m above ground level)	10000	1	/

Note: “/” indicates that no test was conducted. All heights not marked in the table are 1.3m above the ground.

3.4.3 Communication by Taihu Lake in Suzhou suburbs

3.4.3.1 Test environment

Near Taihu Lake in Wujiang, Suzhou, with virtually no obstruction.

1. Module Status:

The test board here is an early evaluation board, not the PAN3029EVB evaluation board provided for customer samples. Customer-provided evaluation boards will perform better than this one.

1) Transmitter: Near Taihu Lake, 0.5m above ground level, slightly obstructed by reeds in front.



Figure 3-30 TX terminal

2) Receiver: Taihu Lake. Due to different environments, taking into account the convenience of testing and minimizing the impact of obstructions, the placement height will be different. There are three placement heights: 0.5m, 1.3m and 2.2m above the ground.



Figure 3-31 RX end (0.5m, 1.3m, 2.2m from left to right)

2. Parameter Configuration:

- 1) Frequency: 499MHz
- 2) Payload Length: 10 Bytes
- 3) SF and BW: There will be some variation at different distances; see test data for details.
- 4) CR: 1
- 5) Output Power: 20dBm

3. Screenshots at Different Distances:

Scenario 1: The transmitter and receiver are on the same side of Taihu Lake, 3810 meters apart. There is some obstruction, but more obstruction than at other locations.

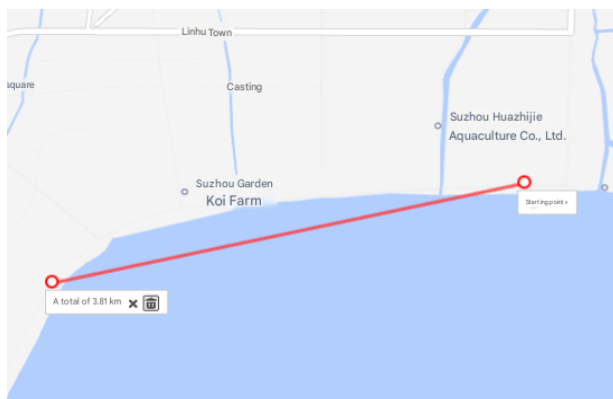


Figure 3-32 Screenshot at a distance of 3810 meters



Figure 3-33 Receiver environment at a distance of 3810 meters

Scenario 2: The transmitter and receiver are on the same side of Taihu Lake, 5090 meters apart, with some obstruction, but not much.

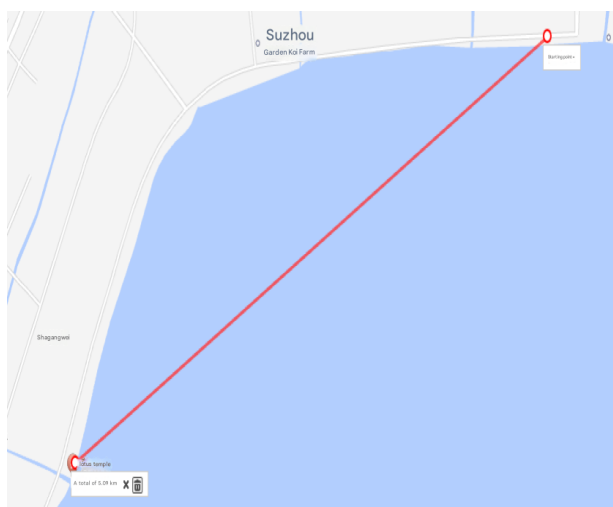


Figure 3-34 Screenshot at a distance of 5090 meters



Figure 3-35 Receiver environment at a distance of 5090 meters

Scenario 3: The transmitter and receiver are on the same side of Taihu Lake, 7,130 meters apart, with some obstruction, but not much.



Figure 3-36 Screenshot at a distance of 7130 meters



Figure 3-40 Receiver environment at a distance of 7130 meters

Scenario 4: The transmitter and receiver are on the same side of Taihu Lake, 9060 meters apart, with some obstruction, but not much.

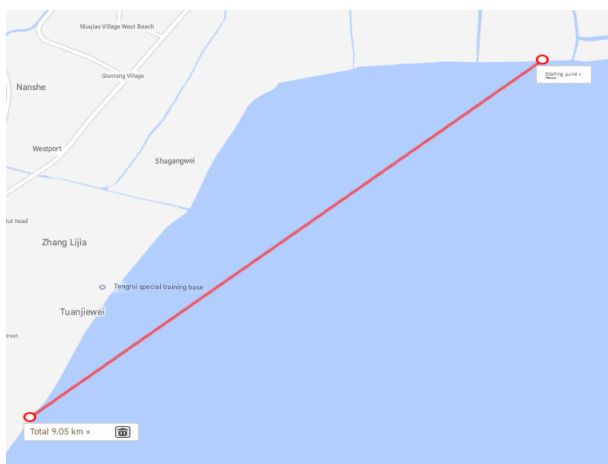


Figure 3-37 Screenshot at a distance of 9060 meters



Figure 3-38 Receiver environment at a distance of 9060 meters

Scenario 5: The transmitter and receiver are on the same side of Taihu Lake, 10,520 meters away. There is some obstruction, but it is minimal.

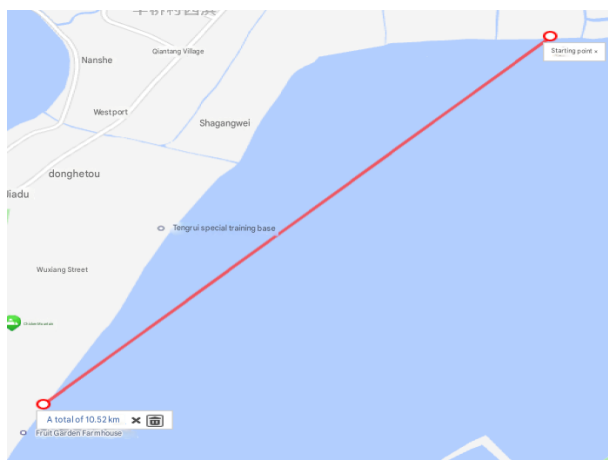


Figure 3-43 Screenshot at a distance of 10,520 meters



Figure 3-44: Receiver environment at a distance of 10,520 meters

Scenario 6: The transmitter and receiver are on the same side of Taihu Lake, 12,880 meters apart. There is some obstruction, but it's minimal.

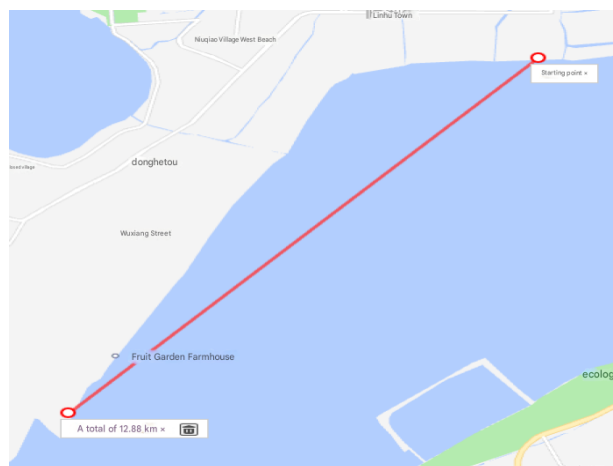


Figure 3-45: Screenshot at a distance of 12,880 meters



Figure 3-46: Receiver environment at a distance of 12,880 meters

Scenario 7: The transmitter and receiver are on opposite sides of Taihu Lake, 14,940 meters apart. There is essentially no obstruction.



Figure 3-47: Screenshot at a distance of 14,940 meters

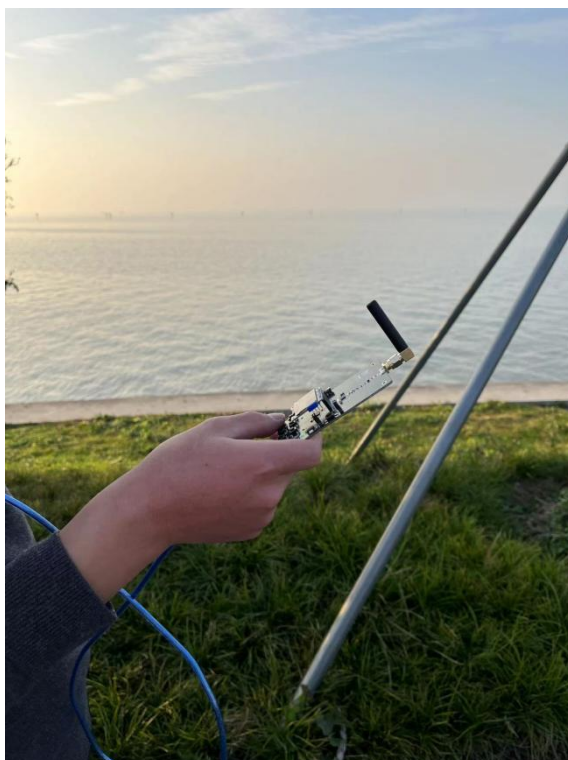


Figure 3-48: Receiver environment at a distance of 14,940 meters

Scenario 8: The transmitter and receiver are on opposite sides of Taihu Lake, 17,420 meters apart. There is essentially no obstruction.



3-49 Screenshot at a distance of 17,420 meters



Figure 3-50 Receiver environment at a distance of 17,420 meters

Scenario 9: The transmitter and receiver are on opposite sides of Taihu Lake, 20,230 meters apart, with virtually no obstruction.



Figure 3-51 Screenshot at a distance of 20,230 meters

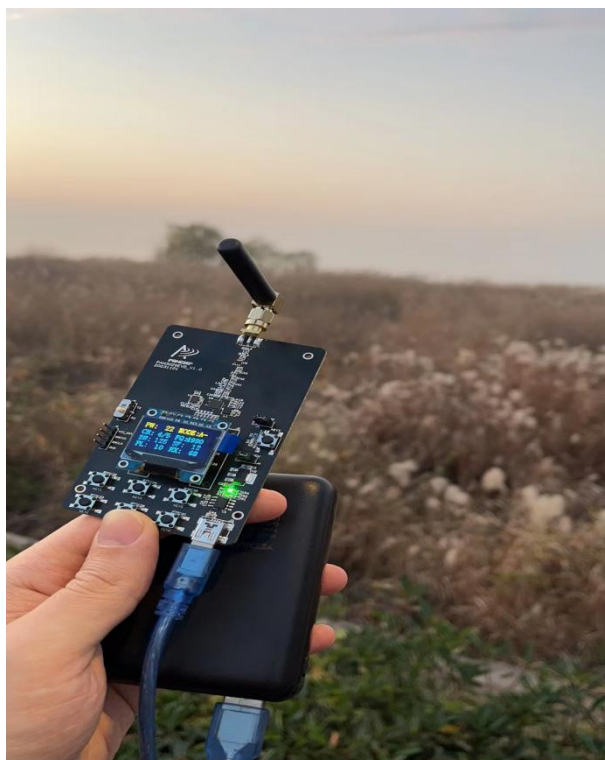


Figure 3-52 Receiver environment at a distance of 20,230 meters

Scenario 10: The transmitter and receiver are on opposite sides of Taihu Lake, 23,100 meters apart, with virtually no obstruction.



Figure 3-53 Screenshot at a distance of 23,100 meters



Figure 3-54 Receiver environment at a distance of 23,100 meters

3.4.3.2 Test steps

1. Use the buttons to switch to TX mode and RX mode, respectively, and configure the corresponding CR/BW/SF parameters.
2. The TX side uses B mode to send 100 packets at a time. The RX side then counts the number of packets received at different distances and under different parameters.

3.4.3.3 Test results

The test results are shown in the table below. At 3810m and 5090m, there is relatively high obstruction, but the 125K SF10 can receive over 80% of the packets. At 7130m and 9160m, packet reception is very good, with the 125K SF10 receiving nearly all packets. Above 10520m, directionality begins to appear. At 17420m and 20230m, directionality is relatively small, resulting in better packet reception. Packets can be received up to 23100m, and no further testing is performed.

Table 3-11 Distance Pull Results Along Taihu Lake in Suzhou

BW (kHz)	SF	Tx end position		Rx end position		Distance (m)	Packet receiving statistics	
							PAN3029	
125	7	200m west from the intersection of Binhu Avenue	Height from the ground 0.5m	300m west of the intersection of Binhu Avenue	Height from the ground 0.5m	3810	4	26
	8						48	55
	9						73	75

	10	and Dongshan Avenue		and Tengfei Road			97	93
125	7			Lotus Temple	Height from the ground 0.5m	5090	0	0
	8		2				0	
	9		62				59	
	10		92				83	
	11		92				99	
500	5			400m southwest from the intersection of Binhu Avenue and Shoucaiyuan Road	Height from ground: 1.3m	7130	0	0
250	5		3				0	
125	5		50				40	
	6		90				57	
	7		73				84	
	8		99				99	
	9		96				100	
	10		98				100	
125	5			300m southwest from the intersection of Binhu Avenue and Shiheshan Road	Height from ground: 1.3m	9060	0	0
	6		1				1	
	7		52				56	
	8		97				99	
	10		100				100	
125	10			Jishan Port Bridge	Height from ground: 1.3m	10520	67	60
	11		88				85	
	12		90				92	
125	9			Beidongshan Branch of Taihu Lake Ecosystem Field Observation Station	Height from ground: 1.3m	12880	11	3
	10		49				57	
	11		92				99	
125	11			150m northwest from the intersection of Nan Taihu Avenue and Guanjie Road	Height from ground: 1.3m	14940	10	/
	12		40				/	
62.5	12		63				/	
125	10			150m north of Xixi Miaogang	Height from ground: 1.3m	17420	6	/
	11		27				98	
	12		70				100	
62.5	12		97				/	
125	10			50m west from the intersection of South Taihu Avenue and Marlboro	Height from ground: 2.2m	20230	61	/
	11		96				/	
	12		100				/	
			99				/	
125	12			60m northwest of the lobby of Taihu Hot Spring Resort Hotel	Height from ground: 1.3m	23100	42	/
62.5	12		59				/	

Note: “/” indicates that no test was performed.

3.4.4 Suzhou Zhongxin Building wall penetration test

3.4.4.1 Test environment

Inside the Suzhou Sino-Singapore Building, the walls are solid walls. The transmitter is on a staircase underground, and the receiver moves in different locations. There are 8 elevators blocking the way between the two staircases.

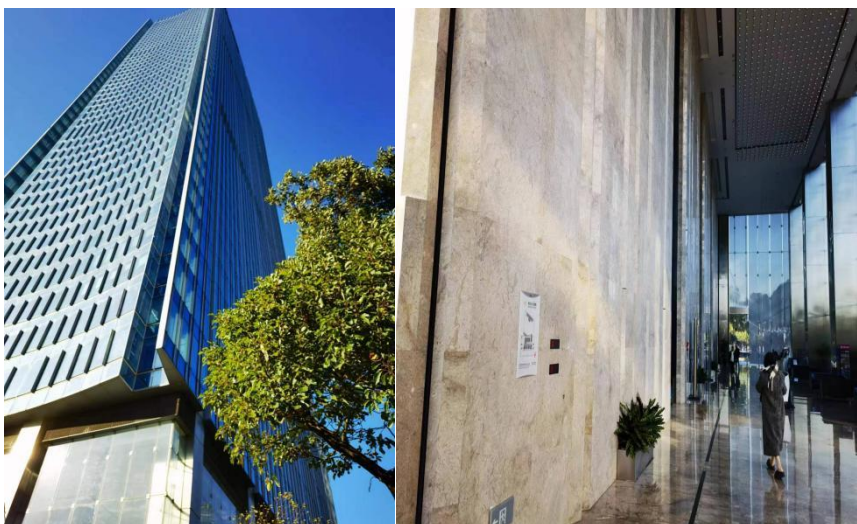


Figure 3-55 Zhongxin Building

1. Module Status:

- 1) Transmitter: Place the transmitter on the first floor underground, on a staircase, 1.4 meters above the ground.



Figure 3-396 TX Side Panorama

- 2) Receiver: Find a corner with minimal obstruction and place the module on each floor, 0.02 meters above the ground.



Figure 3-40 RX Side

2. Parameter Configuration:

- 1) Frequency: 499MHz
- 2) Payload Length: 10Bytes
- 3) SF: 10, BW: 250kHz
- 4) CR: 1
- 5) Output Power: 20dBm

3.4.4.2 Test steps

1. Use the buttons to switch to TX and RX modes, and configure the corresponding CR/BW/SF parameters.
2. The TX sends 100 packets at a time in B mode, and the RX counts the number of packets received at different distances and parameters.

3.4.4.3 Test results

The test results are shown in the table below. In the reinforced concrete structure of the Zhongxin Building, with the receiver located in a corner with significant obstruction, the 125K SF12 can penetrate the wall of the 12th floor. The packet reception rate is over 90%, and the 125K SF11 and SF12 can penetrate the wall of the 11th floor. Because the wall penetration test results are strongly dependent on the environment, the specific wall penetration results are evaluated in combination with the environment.

Table 3-12 Wall penetration distance results for the Suzhou Zhongxin Building

BW (kHz)	SF	Tx end position	Rx end position	Packet receiving statistics	
				PAN3029	
500	5	Basement 1, Zhongxin	4th Floor, Zhongxin Building	99	100

125	5	Building	5th Floor, Zhongxin Building	99	100
250	5			99	99
500	5			61	82
125	5		6th Floor, Zhongxin Building	0	/
	6			0	/
	7			13	16
	8			56	50
	9			90	66
125	10			96	97
	5		7th Floor, Zhongxin Building	0	0
	6			33	13
	7			84	41
	8			90	76
	9			99	100
125	6		9th Floor, Zhongxin Building	2	0
	7			69	22
	8			100	100
	9			97	100
125	9		10th Floor, Zhongxin Building	0	3
	10			28	35
	11			85	94
125	9		11th Floor, Zhongxin Building	1	7
	10			70	66
	11			99	98
	12			99	99
125	9		12th Floor, Zhongxin Building	6	16
	10			66	76
	11			96	64
	12			74	52
125	12		13th Floor, Zhongxin Building	0	0
62.5	12			5	0

Note: The 8th floor is a refuge floor and is not convenient for comparative testing, so it is not tested here.

Revision history

Version	Date	Content
1.0	2023.12	First edition

Document Description

This document may be updated from time to time due to version upgrades or other reasons. Unless otherwise agreed upon, the contents of this document are for guidance only. All statements, information, and recommendations herein do not constitute any warranty, either express or implied.

Trademark

Panchip is a trademark of Panchip Microelectronics Co., Ltd. Other names mentioned in this document are trademarks or registered trademarks of their respective owners.

Disclaimer

All or part of the products, services, or features described in this document may not be within the scope of your purchase or use. Unless otherwise agreed upon in the contract, Panchip Microelectronics Co., Ltd. makes no representations or warranties, express or implied, with respect to the contents of this document.

Contact information



Address: Room 302, Building D, No. 666, Shengxia Road, Zhangjiang Hi-Tech Park, Shanghai



021-50802371

<http://www.panchip.com>

